

Abstract

In this thesis, we focus on improving upon a popular deep reinforcement learning algorithm known as Proximal Policy Optimization (PPO). PPO is an on-policy actor-critic algorithm that has been widely used. One of its core mechanisms - ratio clipping has demonstrated strong performance in various reinforcement learning tasks. We show how ratio clipping bounds the maximum total variation(TV) distance between policy updates, leading to a bound on the approximation error caused by the change in the state distribution.

We then present an inequality that suggests bounding the product of the ratio deviation and its corresponding advantage value. However, simply bounding the product did not result in satisfactory performance. Therefore, we propose a new objective that combines both bounds for improved performance. We call the new algorithm Adaptive clipping PPO(AC-PPO).

AC-PPO is tested on a subset of the 57 Arcade Learning Environments - Atari-5. The experimental results show that AC-PPO outperforms PPO on all five tested games, with smaller KL divergence between policy updates and significantly fewer KL "spikes". On-policy algorithms like PPO typically use only the most recent data to update the policy, discarding the previously collected data as well as old policies. Our results demonstrate the effectiveness of the proposed algorithm in preventing the agent from "overreacting" to signals and unlearning useful policies, a common issue in on-policy algorithms. This study highlights the importance of stable updates in on-policy reinforcement learning algorithms.

As future work, we aim to investigate the reason for failure on the DoubleDunk game, as it may hold key insights for improving our algorithm. We also plan to conduct benchmark testing on other environments. Additionally, we will perform an ablation study to understand the impact of each method on performance, and investigate the algorithm's sensitivity to hyperparameters.